Exploratory Data Analysis Report: Titanic Dataset

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Internship Task 5 Submission

1. Introduction

The goal of this exploratory data analysis is to uncover patterns and insights from the Titanic dataset using statistical and visual exploration techniques. We aim to understand which factors influenced survival.

2. Dataset Overview

Dataset Name: Titanic-Dataset.csv

Total Rows: 891

Total Columns: 12+ (including engineered features)

Key Columns: Survived, Age, Fare, Sex, Pclass, SibSp, Parch, Embarked

Basic Info & Statistics:

- .info() used to identify missing values
- .describe() to observe mean, std, min, max etc.
- .value_counts() for categorical variables like Sex, Embarked, etc.

3. Data Cleaning & Feature Engineering

- Filled missing values in Age and Embarked
- Extracted Title from Name
- Created FamilySize = SibSp + Parch + 1
- Created IsAlone feature

- One-hot encoded Sex, Embarked columns

4. Visual Explorations & Insights

A. Histogram: Age Distribution

Most passengers are between 20-40 years. Few children and elderly.

B. Countplot: Survival by Gender

Female survival rate is significantly higher. Shows 'Women and Children first' pattern.

C. Boxplot: Fare vs Survival

Survivors paid higher fares on average. 1st class had higher chance of survival.

D. Heatmap: Correlation Matrix

Strongest positive correlation with survival: Fare, Pclass. Age has a mild negative correlation.

E. Family Size vs Survival

Medium-sized families had better survival rates. People travelling alone had lower survival.

F. Embarked vs Survival

Most passengers embarked from Southampton (S). Higher survival from Cherbourg (C).

5. Key Findings

- Gender played a key role - females had higher survival.

- Class mattered - higher class = higher survival.

- Fare paid correlates with likelihood of survival.

- Family Size affects survival - small to mid-sized families fared better.

- Children (young age) were prioritized for survival.

6. Conclusion

This EDA helped uncover survival patterns on the Titanic using basic statistics and visualizations. These insights can aid predictive modeling and demonstrate how data storytelling helps in real-world analysis.

7. Tools Used

Python (Pandas, Seaborn, Matplotlib)

Jupyter Notebook

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